



University of Tehran

College of Engineering
School of Electrical & Computer Engineering

Experiment 1

Sessions 1, 2

Clock and Periodic Signal Generation

Digital Logic Laboratory

ECE 045, ECE 895

Laboratory Manual

Spring 1405

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1- Introduction

The goal of this experiment is to introduce the concepts of static characteristics of digital logic gates, propagation delay times, and clock frequency generation using analog oscillator circuits. Understanding these fundamental concepts is essential for digital design, as they determine the speed, timing, and reliability of all sequential digital systems.

In this experiment, you will use the *LTspice* simulation tool to implement and analyze different oscillator circuits for clock generation. You will work with CD4000 series logic gates, the LM555 timer IC, and Schmitt trigger oscillators. By simulating these circuits, you will learn to measure key parameters such as propagation delay, frequency, duty cycle, rise time, and fall time. You will also observe how component values affect oscillator behavior and compare your simulation results with theoretical calculations.

By the end of this experiment, you should have learned:

- Power supply, Function Generator, and Oscilloscope
- CD 4000 Series Basic Logic Gates
- Different oscillator circuits (a LM555 timer IC, Schmitt trigger Oscillator)

2- Clock Generation using ICs and Analog components

In this part, you will understand different methods of clock generation in digital systems. You will use *LTspice* simulation tool for implementing the circuits of this section. Also a separate demo on running examples in *LTspice* will be provided for you.

A- Ring Oscillator

One of the most important parameters in digital logic gates is the propagation delay that is defined as the time from the 50% point of input to the 50% point of output, as shown in figure 1. There are also two more parameters named t_{fall} and t_{rise} that are measured as the time between 10% and 90% point of the signal and vice versa.

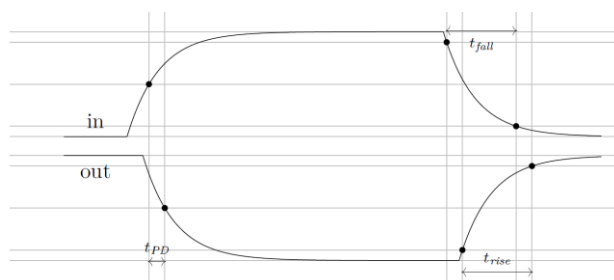


Figure 1. Timing diagram of a logic gate

The delay of logic gates is very small and this imposes a large bandwidth. So, measuring this delay directly using a relatively low-cost oscilloscope may be difficult. An alternative method for measuring this parameter is using a *ring oscillator* as shown in figure 2. A ring oscillator is composed of a chain of odd number of inverters, in which output of last inverter is connected to the input of first one. The time at

which a value feeds back to the same node is the time period of the ring oscillator which equals $2N * Delay_{inv}$, where N is the odd number and $Delay_{inv}$ is the delay of each inverter gate. The delay of each single inverter can be determined by measuring the total delay. Simulate the circuit shown in figure 2 using inverter gate in CD4000 series (CD4049B). Figure 3 shows the schematic of this circuit in LTspice.

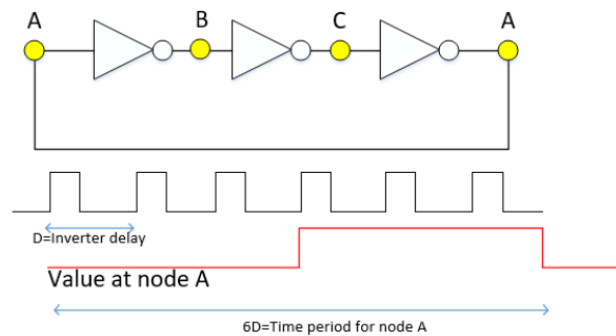


Figure 2. Ring oscillator

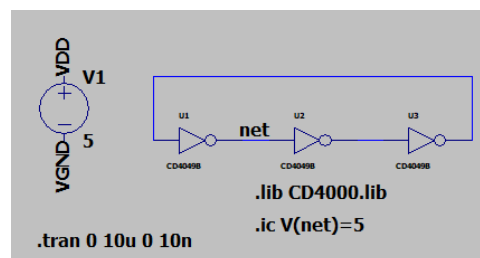


Figure 3. The schematic of ring oscillator in LTspice

Important points to consider when simulating in LTspice:

- Use the CD4049B component for the inverter
- Add a voltage component with DC value set to 5V
- Use label names for nets, including VDD and VGND
- Add the SPICE directive `.lib CD4000.lib` in toolbar:
- Add the initialization directive `.ic V(net)=5` (replace "net" with your actual node name to initiate the node of circuit with corresponded value)

Then perform the following:

1. Measure the propagation delay of the chain by measuring the period time of the output.
2. Measure the delay of a single inverter and compare it with the delay in 7404 TTL specification (Use from 7404 TTL datasheet).

B- LM555 timer

LM555 is among the devices can be used for generating clock signal or time delays. The pin layout of this IC can be seen in figure 4.

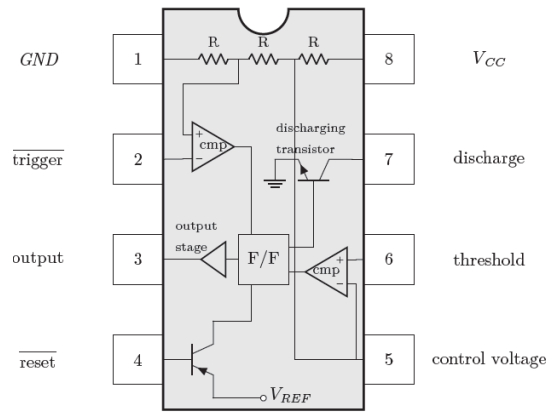


Figure 4. LM555 timer pin-out

This IC operates in three modes: Monostable, Bistable and Astable. The astable mode that we use in this experiment allows the timer to operate as an oscillator that outputs a continuous rectangular pulse of a required frequency. For astable operation, we need two resistors and one capacitor to design a circuit that operates at the frequency required. The timing during which the output is either high or low is determined by these externally connected resistors and capacitors. Note that the durations of the low and high states may be different. Figure 5 illustrates an LM555 configuration for astable mode operation.

The external capacitor $C1$ charges through $R1 + R2$ and discharges through $R2$. Thus, the duty cycle and frequency may be precisely set by selecting the right combination of resistances and capacitance. According to figure 4 and figure 5, the charge time (output high) is given by $T1 = 0.693 * (R1 + R2) * C$ and the discharge time (output low) by $T2 = 0.693 * R2 * C$. Thus, the total time period of square wave is $T = T1 + T2 = 0.693 * (R1 + 2R2) * C$. Consequently, the frequency of oscillation is $\frac{1}{T}$. The duty cycle also can be computed by $\frac{R1+R2}{R1+2R2}$.

These equations describe how we can choose these three values to decide on the frequency and the high and low duration of our signal. With the LM555 timer, the default value of $R1$ in this configuration is $1 \text{ k}\Omega$ and this means that we cannot get a perfect 50% duty cycle. (If we make $R2 \gg R1$ then we can get close.)

Important points to consider when simulating in LTspice:

- The LM555 timer can be found in LTspice under [Misc] folder when selecting a component symbol with **NE555** name
- Add a voltage component with DC value set to 5V
- Use label names VDD of voltage component and  icon in toolbar for ground of voltage component and circuit

Do the following work. Your report must include the procedure you followed, as well as any observation and results.

1. Implement the LM555 in astable mode using the wiring diagram from figure 4 and observe the output. Report the clock frequency and the duty cycle and include the waveform of the output in your report.
2. Change the value of $R2$ resistors to produce different clock frequencies. To do so, $R2$ should be $1 \text{ k}\Omega$, $10 \text{ k}\Omega$ and $100 \text{ k}\Omega$. For this resistor, use a $50 \text{ k}\Omega$ variable resistor. Calculate the frequency and duty cycle using above equations and compare them to the clock signal you see on the output.

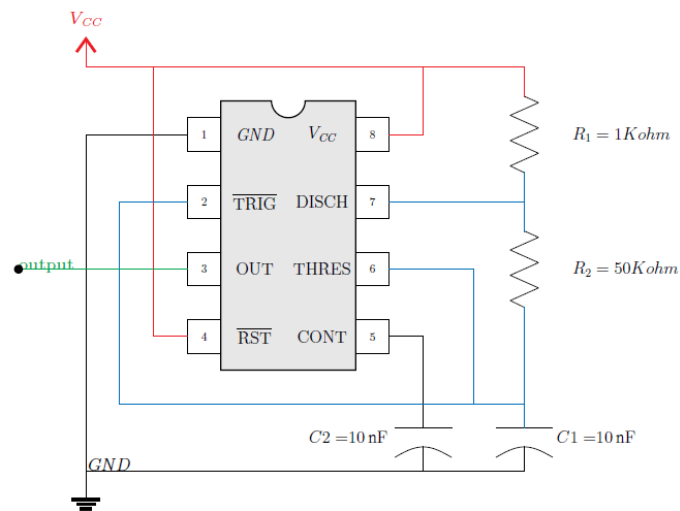


Figure 5. LM555 in astable mode

C- Schmitt Trigger Oscillator

You can see the realization of the Schmitt inverter oscillator in figure 6. In the Schmitt inverter oscillator, $f = \frac{\alpha}{RC}$, where α is a constant. Study the principle of this circuit and be prepared for it.

Important points to consider when simulating in *LTspice*:

- Use the CD40106B component for the Schmitt trigger inverter
- Add a *voltage* component with *DC value* set to 5V
- Use *label names* VDD of voltage component and  icon in toolbar for ground of voltage component and circuit

Then perform the following:

1. Considering the given equation, try the circuit with different values for the resistor and the capacitor and observe the changes. Use 470 Ω , 1 k Ω and 2.2 k Ω for the resistor and 10 nF for the capacitor.
2. Find α parameter.

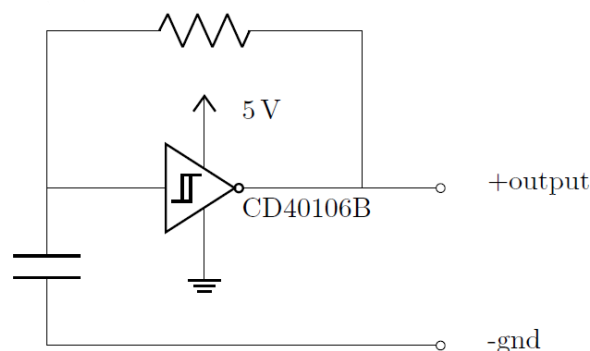


Figure 6. Schmitt inverter oscillator circuit

3- Deliverables

1. Ring Oscillator

- a. Report the observed waveform along with the measured frequency of the chain.
- b. Calculate the delay of a single inverter.
- c. Compare the delay of a single inverter with the delay in 7404 TTL specification.
- d. Upload the *Design1.asc* schematic from *LTspice*.

2. LM555 Timer

- a. For all three cases of R2 resistor(1k, 10k, 100k):
 - i. Include the waveform along with the measured frequency and duty cycle.
 - ii. Using the equations provided in the manual, calculate the expected frequency and duty cycle to verify your measurement.
- b. Explain the changes in the duty cycle by varying the resistor R2.
- c. Upload the LTspice schematic files for variation of R2 resistors: *Design2_1k.asc*, *Design2_10k.asc* and *Design2_100k.asc*.

3. Schmitt Trigger Oscillator

- a. Report the output frequency and calculate the α parameter for all three values of the resistor.
- b. Report the average α .
- c. Upload the LTspice schematic files for variation of R resistors: *Design3_470.asc*, *Design3_1k.asc* and *Design3_2.2k.asc*.